FMA4200 HW3 文杰 123090612

1

(a) Is the model stationary?

Yes.
A moving Average (MA) model of any order q is **always stationary**.

(b) Write down the invertibility condition of the model.

For an $MA(q)$ model to be invertible, the **roots of its characteristic polynomial** must lie outside the unit circle.

For this specific $MA(2)$ process, the characteristic equation is defined as:

$$1 + a_1 z + a_2 z^2 = 0$$

The invertibility condition requires that for all complex roots z that satisfy this equation, their magnitude must be greater than one:

$$|z| > 1$$

This ensures that the MA process can be uniquely represented as an $AR(\infty)$ process.

(c) Find the mean, variance, autocovariance function, and autocorrelation function.

Using the general formulas provided for $MA(q)$ models where $q = 2$:

1. Mean

The expectation of the process is simply the constant term:

$$\mathbb{E}[X_t] = \mu$$

2. Variance

The variance (or the autocovariance at lag 0, γ_0) is:

$$\text{Var}(X_t) = \gamma(0) = \sigma^2(1 + a_1^2 + a_2^2)$$

3. ACVF $\gamma(k) = \text{Cov}(X_t, X_{t-k})$

- **Lag 1:** $\gamma(1) = \sigma^2(a_1 + a_1 a_2)$
- **Lag 2:** $\gamma(2) = \sigma^2 a_2$
- **Lag $k > 2$:** $\gamma(k) = 0$ (The ACVF "cuts off" after lag q)

4. ACF $\rho(k) = \frac{\gamma(k)}{\gamma(0)}$

- **Lag 1:** $\rho(1) = \frac{a_1 + a_1 a_2}{1 + a_1^2 + a_2^2}$
- **Lag 2:** $\rho(2) = \frac{a_2}{1 + a_1^2 + a_2^2}$
- **Lag $k > 2$:** $\rho(k) = 0$

(d)

Please see in my uploaded Jupyter Notebook.

2

(a) Is the model invertible?

Yes.
AR models are always invertible.

(b) Stationary Condition of the AR(2) Process

For an AR(2) model to be stationary, the roots of its **characteristic polynomial** must lie outside the unit circle. Specifically:

The characteristic equation is:

$$1 - b_1 z - b_2 z^2 = 0$$

The **stationary condition** is that both roots z_1, z_2 of this equation must satisfy $|z_i| > 1$ (their magnitudes must be greater than 1).

Explicitly, in terms of the coefficients b_1 and b_2, this corresponds to the following three inequalities:

1. $b_1 + b_2 < 1$
2. $b_2 - b_1 < 1$
3. $|b_2| < 1$

(c) Moments and Correlation Functions

If the AR(2) process is stationary, its statistical properties are derived as follows:

- **Mean ($\mathbb{E}[X_t]$):** The mean μ is given by:

$$\mathbb{E}[X_t] = \mu = \frac{c}{1 - b_1 - b_2}$$

- **ACVF, $\gamma(k)$:** The ACVF satisfies the **Yule-Walker equations**:

$$\gamma(k) = b_1\gamma(k-1) + b_2\gamma(k-2), \quad k \geq 1$$

  To find the specific values, solve the system for $k = 0, 1, 2$:
  - $\gamma(0) = b_1\gamma(1) + b_2\gamma(2) + \sigma^2$
  - $\gamma(1) = b_1\gamma(0) + b_2\gamma(1)$
  - $\gamma(2) = b_1\gamma(1) + b_2\gamma(0)$
- **Variance ($\mathrm{Var}(X_t) = \gamma(0)$):** By solving the system above, the variance is:

$$\gamma(0) = \left(\frac{1 - b_2}{1 + b_2}\right) \frac{\sigma^2}{(1 - b_2)^2 - b_1^2}$$

- **ACF, $\rho(k) = \gamma(k)/\gamma(0)$:** For $k = 1$:

$$\rho(1) = \frac{b_1}{1 - b_2}$$

  For $k \geq 2$, it follows the same recursive relationship as the ACVF:

$$\rho(k) = b_1\rho(k-1) + b_2\rho(k-2)$$

  The ACF of a stationary AR process typically exhibits **exponential decay** and may involve damped oscillations depending on the coefficients.

(d)

Please see in my uploaded Jupyter Notebook.